Alberto Nieto

PRINCIPAL ENGINEER | AI, LLM EVALUATION & AGENTS

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Phoenix, Arizona

PROFESSIONAL SUMMARY

Principal Engineer with 25+ years in financial services, from banking operations to principal-level AI engineering. Builds agentic AI, LLM evaluation and decision models, document and video pipelines, and the cloud infrastructure they run on (GCP, OpenShift/Kubernetes). Outside work, builds open source tools that test what models actually do, then publishes the evidence: 8 Python packages on PyPI. Co-inventor on three pending patent applications. Bilingual in English and Spanish.

WORK EXPERIENCE

Principal Engineer (Executive Director)

Wells Fargo

Builds AI solutions for line-of-business needs, plus the documentation and frameworks that speed developer onboarding and production rollouts.

Phoenix, Arizona

Jul 2025 - Present

- Created AI agent framework to automate line-of-business operations, adopted across multiple teams
- Proposed and built the first fine-tuning of Gemma 3 and Gemma 4 on internal policy documents (October 2026), toward a decision model whose answers are read as next-token probabilities: graphed the corpus into entities and relationships to build the training set, refined it from training logs and sample outputs, and held out data for evaluation. Supported a demo to AI leadership.
- Building decision models for collections and customer-service next-best-action
- Built video processing pipeline handling thousands of videos daily into standard operating procedures for knowledge tree persistence
- Implemented evaluation and evolution frameworks to score and evolve agent definitions for data pipeline tasks (open-sourced as gepa-adk)
- Deploys production workloads to OpenShift (OCP), with Kafka for messaging and Grafana for monitoring
- Built AI workflows whose volume pushed the bank's internal enterprise LLM API to its capacity limits
- Established patterns ensuring non-functional requirements are satisfied with AI deployments
- Audited enterprise API clients for Apigee AI provisioning to enable multi-modal capabilities and proper scaling

Senior Lead Analytics Consultant (Executive Director)

Wells Fargo

Individual contributor and the team's go-to engineer for GenAI and agentic AI frameworks. Focused on proof of concepts and production-ready AI implementations.

Phoenix, Arizona

Jun 2022 - Jul 2025

- Architected scalable OCR pipeline processing 500K+ documents using Gemini, with results stored in Teradata and SQLite for distributed processing
- Developed intelligent agents for database querying, KPI summarization, SQL validation, and structured QA using Python and Google ADK
- Built reusable data pipelines for document ingestion, OCR, and validation using Selenium, HTTPX, and Docling with Gradio/Streamlit frontends
- Designed modular, scalable architecture patterns reused across multiple teams and projects
- Implemented git-based dev workflow with peer reviews, pull requests, and release standards to mature end-user compute tool development

Lead Analytics Consultant (AVP)

Wells Fargo

T-shaped role providing leadership, oversight, and direct development in ETL, visualization, and advanced analytics workstreams.

Phoenix, Arizona

Jan 2018 - Jun 2022

- Onboarded applications to Pivotal Cloud Foundry with Splunk logging and Blackduck security scans
- Built web scrape automation for business intelligence visualization previews and validation
- Created dynamic databases for manual and automated data ingestion from flat-file data providers
- Engaged technology peer groups to execute high-priority regulatory data analytics

Applications Systems Engineer (AVP)

Wells Fargo

Team lead for vulnerability business intelligence development team.

Phoenix, Arizona

Nov 2016 - Jan 2018

- Provided guidance to development team and business partners on BI solutions for emergency remediation and reporting
- Consulted leadership on data architecture roadmap for vulnerability remediation
- Created ad-hoc SQL reports across systems of record to illustrate compliance status

Business Systems Consultant (AVP)

Wells Fargo

Created business intelligence reporting and tools for regulatory compliance and leadership decision-making.

Phoenix, Arizona

Oct 2009 - Nov 2016

- Built and supported ongoing asset validation processes
- Created compliance metrics for server footprint tracking
- Developed business process reports for server implementation, changes, and decommission time-to-market

Technology Manager

Wells Fargo

Led team of technical professionals supporting end-user technology across four administrative sites.

Phoenix, Arizona

2006 - 2009

- Led 8-9 technicians across four administrative sites
- Led desktop OS migrations and provided onsite hardware swap support
- Expedited Business Continuity Plan documentation to 100% compliance for disaster recovery across four business lines
- Led monthly compliance efforts for security patching, encryption, and asset tracking

Team Lead / Leadership Development Program / Operations

Wells Fargo / Bank of America

Progressive career from teller to operations analyst to leadership development program, building deep financial services domain expertise.

Phoenix, Arizona

1999 - 2006

- Selected for Wells Fargo Leadership Development Program (2004-2005), rotating through operations and technology departments
- Built and presented a SharePoint hosting solution
- Led six technicians at two admin sites with desktop end-user support
- Progressed through teller, personal banker, lead teller, operations processor, and operations analyst roles (1999-2004)

PROJECTS

gepa-adk

Evolutionary prompt optimization for AI agents using Google ADK. Automatically evolves agent instructions through iterative improvement. Async-first engine with hexagonal architecture and protocol-based interfaces. Apache-2.0.

adk-secure-sessions

Encrypted session storage for Google ADK agents. Drop-in replacement using Fernet encryption with PBKDF2 key derivation. Apache-2.0.

docvet

CLI tool for Python docstring quality vetting. 32 rules across five checks (presence, enrichment, freshness, griffe, mkdocs coverage), including git-based staleness detection via diff and blame. Production/Stable. MIT.

vramfit

Hardware-targeted LLM quantization. Measures per-layer compression damage, then solves a mixed-precision recipe against a real VRAM budget (scan to plan to pack). Fit Llama-3.3-Nemotron-Super-49B onto a single 24 GiB RTX 4090: 20.36 GiB packed against a 20.47 GiB weight budget, beating every in-budget baseline on KL divergence. Three packs published across three model families and 24/16 GiB budgets, with per-layer sensitivity maps for the Nemotron packs published for independent reproduction. MIT.

saucier

Provenance-first extraction: parses Escoffier's January 1909 edition into a catalogue of 151 preparations where every claim traces to a source line, with no model in the loop. Publishes the deterministic baseline (94 of 151 state no base) that a later agentic system must beat. 8 ADRs, Diátaxis documentation, API reference generated from docstrings. MIT.

turboquant-vllm

First TurboQuant vLLM plugin, built paper-to-PyPI in 72 hours after Google's ICLR 2026 publication. 3.76x KV cache compression across 7 model families; fused paged kernels cut memory traffic 8.5x.

typevet

Asks an open model typed questions about text and images and reads each answer as a probability from its next-token distribution; runs Gemma 4 31B on a 24 GiB card (llama.cpp or vLLM). With a receipt photo attached it caught 6 of 6 swapped totals that Gemma, reading text alone, had passed as "match" at ≥99.96% confidence. A pre-registered calibration study (16 series, decision rule and split fixed before any fit) found isotonic recalibration sufficient on 5 of 6 decision series, so it recommended against fine-tuning. MIT.

judgevet

Typed Python client, CLI and MCP server for TypeSafe's Jev judgment model, with a published calibration study: Jev's scam-text calibration gap measured at 7 points (ECE 0.073) with sorting score 0.995. Ships a verification ledger that separates live-verified from inferred API claims, checked by a dedicated auditing agent. MIT.

EDUCATION

Bachelor of Science, Accounting Information Systems
DeVry University

AWARDS & HONORS

Named as co-inventor on a second and third pending patent application 2026

Named as co-inventor on a pending patent application January 2026

Top Performer 2018 (New Orleans) - Recognition for vulnerability remediation efforts

Top Performer 2014 (Nashville) - Recognition for asset reporting initiatives

SKILLS

Core Competencies: Generative AI & Agents (Google ADK, Pydantic-AI), Python, Cloud Engineering (GCP, OpenShift/Kubernetes, PCF), Streaming & Observability (Kafka, Grafana), Multi-agent engineering workflows (specifier to builder to independent reviewer), Data Engineering & Pipelines, OCR & Document Intelligence, Video Inference, CI/CD & DevOps, Enterprise Architecture

Languages: English (native), Spanish (bilingual)